UNCLASSIFIED

HOME

MSS Training Curriculum — USAREUR-AF

Version 3.0 • March 2026

i NEW TO MSS?

Start with the Quick Start guide before reading TM-10. It gets you operational in 30 minutes: log in, navigate, filter data, and export a view. → **QUICK_START.pdf**

▶ BLUF

Find your level below and follow the training path. All personnel start at TM-10. Builders add TM-20. Data specialists continue to TM-30. Technical roles proceed to the appropriate TM-40 track. Use the sidebar to navigate to your specific guidance.

STEP 1 — FIND YOUR LEVEL

YOU ARE…	START HERE	THEN	TAB
Any Soldier, officer, or civilian — using MSS to access and consume data	**TM-10 (User)**	—	TM-10
All staff — building dashboards, forms, or pipelines without coding	**TM-10**	TM-20 (Builder)	TM-20
Data-adjacent specialist — 17/25-series, G2, data analyst	**TM-10 → TM-20**	TM-30 (Advanced)	TM-30
Technical specialist — ORSA, AI/ML, MLE, PM, KM, SWE roles	**TM-10 → TM-20 → TM-30**	TM-40 (track)	TM-40
Senior leader (O-5+ / SGM+) directing a data-capable formation	**Data Literacy (SL)**	Optional: Data Literacy	DOCTRINE
Anyone wanting data literacy background before touching MSS	**Data Literacy**	TM-10	DOCTRINE

STEP 2 — FOLLOW THE TRAINING PATH

A

DATA LIT (SL) — OPTIONAL (O-5+ / SGM+)

Data Literacy for Senior Leaders

Principles, command responsibilities, decision frameworks

B

DATA LITERACY — RECOMMENDED (ALL PERSONNEL)

Data Literacy Reference

Platform-agnostic data literacy; recommended before TM-10

1

TM-10 — REQUIRED FOR ALL PERSONNEL

Maven User Manual

CAC login, navigation, Workshop apps, data viewing, AI tools, security

2

TM-20 — ALL STAFF (NO-CODE BUILDER)

Builder Manual

Pipeline Builder (visual), Ontology Manager UI, Workshop app builder

3

TM-30 — DATA-ADJACENT SPECIALISTS

Advanced Builder Manual

Complex app design, Ontology architecture, governance, C2DAO standards

4

TM-40G–L — TECHNICAL SPECIALIST TRACKS

Developer Manuals — 6 Tracks by Role

40G (ORSA) • 40H (AI Eng) • 40I (MLE) • 40J (PM) • 40K (KM) • 40L (SWE) — Advanced versions at TM-50G–L

i TM-40 TRACK SELECTION — BY MOS / ROLE

ORSA (TM-40G): FA49, G2/S2 quantitative analysts, wargame specialists
AI Engineer (TM-40H): AI/ML specialists, 17A/17C
ML Engineer (TM-40I): ML engineers, data scientists (GS/contractor)
Program Manager (TM-40J): G8/S8, PMs, resource managers
Knowledge Manager (TM-40K): KMOs, 37F, institutional memory leads
Software Engineer (TM-40L): 17A/17C, 25D/25U, GS/contractor SWEs
MOS not listed? Ask your commander — you may need TM-10 only, or be assigned a track based on unit needs.

Full MOS-to-track mapping: **Specialist Tracks →**

• NOTE — MSS ACCOUNT ACCESS

MSS access requires a provisioned account. Contact your unit data steward to initiate a request. Allow 3–5 business days for provisioning. Do not wait until the start of a deployment or exercise.

ADVANCED & TECHNICAL TRACKS

Specialist Tracks — TM-40 & TM-50 Series

Role-specific developer manuals for ORSA, AI Engineer, ML Engineer, Program Manager, Knowledge Manager, and Software Engineer. Prerequisite: TM-30 complete.

Access Specialist Tracks →

TM-10

Maven User Manual — All Personnel

No technical background required • Prerequisite: None

▶ BLUF

TM-10 teaches you how to log in, navigate MSS, read dashboards, submit data through forms, use AI tools, and stay within authorized boundaries. Required for all USAREUR-AF personnel before operating MSS. **Request your account before your first class — provisioning takes 3–5 business days.**

⚑ CAUTION — REQUEST YOUR ACCOUNT FIRST

You cannot log in without a provisioned account. **Do not wait until the first day of class.** Request your account from your unit data steward now — allow 3–5 business days. Steps are in Section 1 below.

1. GETTING ACCESS — DO THIS FIRST

1. Find your **unit data steward** (your data steward is typically your unit's S-1, S-2, or IT section — ask your squad leader if you're unsure).
2. Ask them to submit an MSS account request with your name, unit, MOS, and required access level.
3. MSS admin team provisions your account and assigns **markings** (markings = the data categories and classification levels you're authorized to see).
4. You receive notification when account is active — typically 3–5 business days.
5. Receive the MSS portal URL from your unit data steward. This is your login link.

• NOTE — DATA STEWARD

Your **unit data steward** manages MSS access for your unit. They are your first point of contact for account requests, access problems, and data errors. If you don't know who they are, ask your squad leader or check with your unit S-1.

2. WHAT IS MSS?

MSS is a secure, web-based platform where your unit's data lives and can be analyzed and acted upon. Think of it as a shared operations center for data: information from logistics, personnel, readiness, and other systems is collected, organized, and made accessible through applications your unit uses every day.

MSS is built on the Palantir Foundry platform, authorized for Army use under the Maven Smart System program.

• NOTE — WHAT IS FOUNDRY?

Palantir Foundry is a commercial data platform that Army headquarters selected to run MSS. You do not need to know how it works — just how to use it. The word "Foundry" may appear in help documentation and system menus; it refers to the same platform as MSS.

MSS DOES

- Stores data from Army systems in a single, organized location
- Makes data visible through applications and dashboards
- Enables units to update records, report status, and track readiness
- Provides analysis tools for authorized personnel
- Supports AI-assisted analysis through authorized tools

MSS IS NOT

- Not a replacement for official systems of record (DCPDS, GCSS-A, MEDPROS, etc.)
- Not classified by default — classification depends on data markings
- Not a public system — access is controlled and audited

USAREUR-AF MISSION AREAS

- **Personnel Readiness** — soldier readiness status
- **Logistics** — equipment availability & maintenance
- **Operational Reporting** — SITREPs and updates
- **Planning** — orders, unit positions, task org
- **C2** — unit status across the AOR

3. SECURITY RESPONSIBILITIES

⚠ WARNING

Unauthorized access to, disclosure of, or modification of data in MSS may constitute a violation of 18 U.S.C. § 1030 (Computer Fraud and Abuse Act) and applicable Army regulations. Violations may result in disciplinary action, loss of access, and criminal prosecution.

1. Use only your own credentials. Do not share your CAC, PIN, or access tokens.
2. Access only data you are authorized to view.
3. Report misrouted data immediately. If you see data at a higher classification than your clearance — STOP and report it.
4. Do not export data without authorization. Exports are logged.
5. Log out when done. Do not leave an MSS session unattended on an unlocked workstation.
6. Report security incidents immediately to your supervisor and unit security officer.

4. TASKS

TASK: ACCESS THE MAVEN SMART SYSTEM

CONDITIONS	STANDARDS	EQUIPMENT
Provisioned MSS account, CAC reader, approved workstation and browser	Successfully authenticate with CAC and reach the MSS home screen	CAC, CAC reader, workstation, MSS portal URL (from unit data steward)

1. Insert your CAC into the CAC reader.
2. Open an approved web browser (Chrome or Firefox recommended).
3. Navigate to the MSS portal URL provided by your unit data steward.
4. When prompted, select your **authentication certificate** (not email certificate).
5. Enter your CAC PIN when prompted.
6. MSS home screen loads — you are now logged in.

> ⚑ CAUTION
>
> Do not save your PIN in the browser. Do not allow the browser to remember your login. MSS sessions may contain sensitive information.

TASK: NAVIGATE THE MSS HOME SCREEN

CONDITIONS	STANDARDS
Logged into MSS	Identify all major navigation elements; locate search, notifications, and profile

ELEMENT	LOCATION	PURPOSE
Search bar	Top center	Find datasets, applications, and projects
Notification bell	Top right	System alerts, workflow updates
User profile icon	Top right	Account settings, markings (your authorized data categories), logout
Compass (file explorer)	Left sidebar	Browse all MSS resources
Home button (logo)	Top left	Return to home screen from anywhere
Pinned items	Home main area	Shortcuts to frequently used resources
Recent activity	Home main area	Recently visited datasets and apps

5. REPORTING PROBLEMS

PROBLEM TYPE	WHO TO CONTACT
Cannot log in	MSS Help Desk
Cannot access a project	Unit data steward
Data appears incorrect	Unit data steward (do not correct it yourself)
System error or crash	MSS Help Desk (provide error code and screenshot)
Security incident	Supervisor and unit security officer — IMMEDIATELY
Application not working	MSS Help Desk

SCHED UPCOMING TRAINING — TM-10

• ENROLLMENT

Contact the listed POC to reserve a seat. Bring your CAC and ensure your MSS account request is submitted at least 5 business days before the course start date. Virtual sessions require MS Teams access and a headset.

DATES	LOCATION	FORMAT	POC	SEATS	STATUS
14–15 APR 2026	Wiesbaden, Clay Kaserne, Bldg 3312, Rm 104	In-Person	SSG Johnson	20 / 20	**OPEN**
05–06 MAY 2026	Grafenwöhr, Bldg 244, Conf Rm B	In-Person	SFC Davis	8 / 20	**8 SEATS REMAINING**
18–19 JUN 2026	Stuttgart, Kelley Bks, Bldg 3357	In-Person	SSG Martinez	20 / 20	**OPEN**
09–10 JUL 2026	Virtual (MS Teams)	Virtual	SSG Johnson	30 / 30	**OPEN**

Duration: 2 days (16 hours). Course runs 0800–1700 each day. All dates subject to change — confirm with POC 5 days prior.

TM-20

No-Code Builder Manual — All Staff

Prerequisite: TM-10 • No coding required

▶ BLUF

TM-20 teaches you how to ingest data, build Workshop applications, create Object Types and basic Actions, and manage projects — all using the graphical user interface. No coding required. Prerequisite: TM-10 complete.

COMPETENCIES UPON COMPLETION

PROJECT MANAGEMENT (UI)

- Create and organize Foundry projects via the UI
- Set up folder structure: raw / staging / curated layers
- Manage project access and permissions via UI
- Follow USAREUR-AF naming conventions and builder standards

DATA INGEST (NO CODE)

- Ingest data using Pipeline Builder — visual, no code
- Configure connectors and file sources via UI
- Schedule pipeline runs via UI
- Understand raw / staging / curated dataset layers

ONTOLOGY (UI)

- Create Object Types and set primary keys via Ontology Manager UI
- Create Link Types between objects via UI
- Build basic Actions via Action Editor UI
- Test Object Types in Object Explorer before building apps

WORKSHOP APPLICATIONS

- Build and publish Workshop applications with dashboards, forms, and filters
- Select and configure appropriate widgets for each use case
- Apply access controls and publish to users

BRANCHING & GOVERNANCE

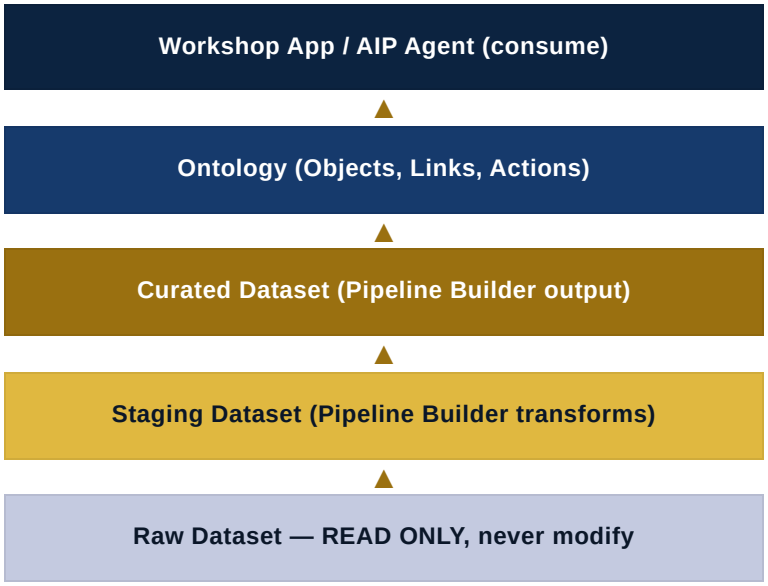
- Use Foundry branching to build and promote via UI
- Distinguish development from production environments

- Apply USAREUR-AF builder standards

*Branching = making a test copy of your work before going live. You build in the **dev branch** (your sandbox), test it, then publish to **production** (what users see).*

THE FOUNDRY DATA STACK

Data flows through layers. As a TM-20 builder, you work in the middle layers using visual tools. Never modify raw data — report data errors to your data steward instead.



WORKSHOP WIDGET SELECTION

YOU NEED TO…	USE THIS WIDGET
Display many objects or records	Object Table
Show details for one selected object	Object Detail
Let users filter the data they see	Filter Panel / Dropdown
Show a chart (bar, line, pie)	Chart Widget
Show geographic data on a map	Map Widget
Let users write or update data	Button + Action or Action Form

YOU NEED TO…	USE THIS WIDGET
Show a single key metric prominently	Metric Tile
Navigate between app sections	Navigation / Tab Widget

ONTOLOGY SETUP ORDER (UI STEPS)

1. Confirm curated dataset exists and is populated (Pipeline Builder pipeline passing)
2. Open **Ontology Manager** in the left sidebar
3. Create Object Type → set primary key property → map properties from curated dataset
4. Create Link Types between related Object Types (if needed)
5. Publish ontology branch and test in **Object Explorer**
6. Build Workshop app only after Object Explorer confirms objects are visible

NAMING CONVENTIONS

OBJECT TYPE	CONVENTION	EXAMPLE
Datasets (path)	`/Project/AOR/source/raw|staging|curated`	`/USAREUR/EUR/personnel/c`
Object Types	`PascalCase`	`UnitStatus`
Properties (API name)	`camelCase`	`unitName`
Properties (display name)	Title Case	`Unit Name`
Link Types	`PascalCase` verb form	`HasEquipment`
Workshop app names	Unit + function + version	`EUR-Personnel-Readiness-`

⚑ CAUTION — SHARED RESOURCES

Changes to shared datasets and Object Types affect all downstream applications and users. Before modifying a shared resource, coordinate with your data steward.

SCHED UPCOMING TRAINING — TM-20

• ENROLLMENT

TM-10 must be complete before attending TM-20. Contact the listed POC to reserve a seat. Bring completion certificate from TM-10 on day one.

DATES	LOCATION	FORMAT	POC	SEATS	STATUS
21–23 APR 2026	Wiesbaden, Clay Kaserne, Bldg 3312, Rm 104	In-Person	SFC Chen	7 / 15	**7 SEATS REMAINING**
12–14 MAY 2026	Grafenwöhr, Bldg 244, Conf Rm B	In-Person	SSG Williams	3 / 15	**3 SEATS REMAINING**
22–24 JUN 2026	Stuttgart, Kelley Bks, Bldg 3357	In-Person	SFC Chen	15 / 15	**OPEN**
14–16 JUL 2026	Virtual (MS Teams)	Virtual	SSG Williams	20 / 20	**OPEN**

Duration: 3 days (24 hours). Course runs 0800–1700 each day. Prerequisite: TM-10 complete. All dates subject to change — confirm with POC 5 days prior.

TM-30

Advanced Builder Manual — Data-Adjacent Specialists

Prerequisite: TM-10 + TM-20 • 17/25-series • G2 • Data analysts

▶ BLUF

TM-30 is for personnel who design and own MSS solutions. This level covers complex application design, advanced Pipeline Builder, Ontology architecture, AIP Logic configuration, data governance, and C2DAO standards. All work is done via the UI; coding is escalated to TM-40 developers.

⚑ CAUTION

Modifications to shared datasets or Object Types at TM-30 level affect all downstream applications and users across the formation, including coalition partners. Coordinate with your unit data steward and the USAREUR-AF C2DAO before publishing any changes to production resources.

COMPETENCIES UPON COMPLETION

ADVANCED WORKSHOP DESIGN

- Design multi-page Workshop applications with conditional logic and variable passing
- Build dynamic layouts: show/hide panels based on user selections
- Design navigation flows and inter-page parameter handoff
- Publish and manage application versions

ADVANCED PIPELINE BUILDER

- Build multi-source join pipelines with complex aggregations (visual)
- Design scheduled and triggered pipeline runs
- Review and interpret data lineage graphs
- Escalate to TM-40 when code transforms are required

ONTOLOGY ARCHITECTURE

- Design Object Type and Link Type models via Ontology Manager UI
- Architecture thinking: model for downstream app requirements, not just source data

ADVANCED ANALYSIS

- Conduct advanced Contour analysis: complex aggregations, pivots, calculated columns, saved views
- Build advanced Quiver dashboards with multi-object analysis and linked views

- Design Action workflows with validation and approval logic via UI
- Coordinate ontology changes with all downstream application owners

- Create reusable analysis templates for unit use

AIP LOGIC & CONFIGURATION

- Configure AIP Logic workflows via the UI (not author them — that is TM-40H)
- Connect AIP agents to Workshop applications
- Review and validate AI workflow outputs before production deployment

GOVERNANCE & PRODUCTION

- Manage branching and production promotion via UI
- Apply USAREUR-AF C2DAO governance standards and naming conventions
- Manage governance workflows with data stewards
- Ensure coalition-facing products have C2DAO coordination and NAFv4 compliance review

TM-30 vs TM-40 — WHAT YOU OWN VS WHAT YOU ESCALATE

YOU OWN AT TM-30 (UI)	ESCALATE TO TM-40 WHEN…
Application design and UX	Custom Python/PySpark transforms needed
Ontology model design (via UI)	Functions on Objects (TypeScript) required
Advanced Pipeline Builder (visual)	Incremental watermark or code logic needed
AIP Logic configuration	Custom AIP Logic workflow authoring needed
Governance coordination	External application (OSDK) needed
Production promotion via UI	CI/CD pipeline automation needed

C2DAO GOVERNANCE GATES — HARD STOPS

REQUIREMENT	TM-30 ACTION	HARD GATE?
New shared Object Type or dataset	Coordinate with C2DAO before publishing to production	Yes

REQUIREMENT	TM-30 ACTION	HARD GATE?
Coalition / MPE-facing data product	C2DAO coordination + NAFv4 compliance review	Yes — do not skip
Schema change to existing shared resource	Notify all downstream owners; coordinate with steward	Yes
New AIP Logic workflow on operational data	Authorization review before deployment	Yes
Access permission changes	Submit through formal request to unit data steward	Yes

SCHED UPCOMING TRAINING — TM-30

• ENROLLMENT

TM-10 and TM-20 must be complete before attending TM-30. Class size is limited. Contact POC early — seats fill quickly. Bring TM-10 and TM-20 completion certificates on day one.

DATES	LOCATION	FORMAT	POC	SEATS	STATUS
28 APR – 02 MAY 2026	Wiesbaden, Clay Kaserne, Bldg 3312, Rm 104	In-Person	CW3 Thompson	4 / 10	**4 SEATS REMAINING**
15–19 JUN 2026	Virtual (MS Teams)	Virtual	CW2 Rodriguez	15 / 15	**OPEN**
17–21 AUG 2026	Wiesbaden, Clay Kaserne, Bldg 3312, Rm 104	In-Person	CW3 Thompson	10 / 10	**OPEN**

Duration: 5 days (40 hours). Course runs 0800–1700 each day. Prerequisites: TM-10 and TM-20 complete. All dates subject to change — confirm with POC 5 days prior.

NEXT LEVEL — AFTER TM-30

Specialist Tracks — TM-40 & TM-50 Series

Six role-specific tracks for ORSA, AI Engineer, ML Engineer, Program Manager, Knowledge Manager, and Software Engineer. Select your track based on MOS/role.

Access Specialist Tracks →

SPECIALIST TRACKS

TM-40 & TM-50 Series — Advanced Technical Manuals

Prerequisite: TM-30 complete • Six tracks by role

▶ BLUF

Specialist tracks are for personnel who build, engineer, and maintain MSS solutions at the technical level. Select your track by MOS/role below. All tracks require TM-30 as a prerequisite. Advanced versions (TM-50) are available for personnel who have completed the corresponding TM-40 track.

TRACK SELECTION BY MOS / ROLE

ROLE / MOS	RECOMMENDED TRACK	ADVANCED
FA49 — Operations Research Analyst	**TM-40G (ORSA)**	TM-50G
G2/S2 quantitative analyst	**TM-40G (ORSA)** or TM-40K (KM)	TM-50G / TM-50K
17A/17C — Cyber officer/NCO	**TM-40L (SWE)** or TM-40H (AI Eng)	TM-50L / TM-50H
25D — IT specialist	**TM-40L (SWE)**	TM-50L
AI/ML engineer (GS/contractor)	**TM-40H (AI Eng)** or TM-40I (MLE)	TM-50H / TM-50I
Data scientist (GS/contractor)	**TM-40G (ORSA)** or TM-40I (MLE)	TM-50G / TM-50I
G8/S8 — Resource manager	**TM-40J (PM)**	TM-50J
Program Manager (PM / GS)	**TM-40J (PM)**	TM-50J
KMO / Knowledge Officer / 37F	**TM-40K (KM)**	TM-50K
G9 / Civil Affairs	**TM-40J (PM)** or TM-40K (KM)	TM-50J / TM-50K

TM-40 SERIES — SPECIALIST TRACKS (Level 1)

TM-40G — ORSA TM-30 REQ.

Operations Research & Systems Analysis

FA49 • G2/S2 quant analysts • Wargame analysts

PREREQ: TM-30 • ADVANCED: TM-50G • View TM-40 Series →

TM-40H — AI Engineer

TM-30 REQ.

AIP Logic, Agent Studio & LLM Integration

AI/ML specialists • 17A/17C

PREREQ: TM-30 • ADVANCED: TM-50H • View TM-40 Series →

TM-40I — ML Engineer

TM-30 REQ.

Model Development, Validation & Deployment

ML engineers • Data scientists (GS/contractor)

PREREQ: TM-30 • ADVANCED: TM-50I • View TM-40 Series →

TM-40J — Program Manager

TM-30 REQ.

Pipeline Mgmt, Milestones & Portfolio Health

G8/S8 • PMs • G9/Civil Affairs

PREREQ: TM-30 • ADVANCED: TM-50J • View TM-40 Series →

TM-40K — Knowledge Manager

TM-30 REQ.

Forms, Lessons Learned & Institutional Memory

KMOs • 37F • G9 / Civil Affairs

PREREQ: TM-30 • ADVANCED: TM-50K • View TM-40 Series →

TM-40L — Software Engineer

TM-30 REQ.

Python/TypeScript, OSDK & Code Transforms

17A/17C • 25D • GS/contractor SWEs

PREREQ: TM-30 • ADVANCED: TM-50L • View TM-40 Series →

TM-50 SERIES — ADVANCED TRACKS (Level 2)

i NOTE

TM-50 tracks are advanced-level continuations requiring the corresponding TM-40 track as a prerequisite. They cover expert-level techniques, production architecture, and operational integration at scale.

Open TM-50 Series Full Reference →

TM-40

Technical Specialist Tracks — Developer Manuals

Prerequisite: TM-30 • Six tracks by role • Advanced versions at TM-50

► BLUF

The TM-40 series covers developer-level capabilities requiring coding, advanced tooling, or specialized technical expertise. Six specialist tracks (TM-40G–L) target specific technical roles; six warfighting function tracks (TM-40A–F) are available by functional area. Prerequisite for all tracks is TM-30. Select your track using the MOS/Role table on the HOME tab.

TM-40G — ORSA Track

TM-30 REQUIRED

Operations Research & Systems Analysis

FA49 • G2/S2 quantitative analysts • Wargame analysts

- Configure Code Workspaces (Python/R) within Foundry
- Statistical modeling: regression, classification, validation for readiness/logistics
- Time series forecasting with ARIMA/SARIMA patterns
- Monte Carlo simulation for COA comparison and risk quantification
- Linear programming for resource allocation and scheduling optimization
- Wargame/exercise data collection architecture and aggregation pipelines
- Analytical decision support products (Quiver/Contour) to commander standard
- Communicate uncertainty: confidence intervals, sensitivity analysis, briefing standards

PREREQ: TM-30 • ADVANCED: TM-50G

TM-40H — AI Engineer Track

TM-30 REQUIRED

AIP Logic, Agent Studio & LLM Integration

AI/ML specialists • 17A/17C

- Author AIP Logic workflows: prompt engineering, chain design, output handling
- Build and configure AIP Agent Studio agents with tools, memory, and orchestration
- Implement LLM integration patterns: ontology data grounding, RAG, context construction
- Apply AI safety requirements: human-in-the-loop gates, output validation, OPSEC
- Write Python transforms that prepare data for AI consumption
- Connect AIP Logic workflows to Object Types and Actions
- Test and red-team AI outputs; evaluate quality against defined standards

- Deploy and monitor AIP Logic workflows in production

PREREQ: TM-30 • ADVANCED: TM-50H

TM-40I — ML Engineer Track

TM-30 REQUIRED

Code Workspaces, Model Training & Deployment

ML engineers • Data scientists building/deploying models on MSS

- Configure Code Workspaces for model development (GPU, packages, environment management)
- Build and evaluate ML models within the Foundry environment
- Manage model versioning, experiment tracking, and reproducibility
- Deploy models to production and integrate with Ontology Objects and Actions
- Implement MLOps patterns: monitoring, drift detection, retraining triggers
- Apply responsible AI practices and model documentation standards for operational use

PREREQ: TM-30 • ADVANCED: TM-50I

TM-40J — Program Manager Track

TM-30 REQUIRED

PM Dashboards, Portfolio Analysis & Reporting Pipelines

PMs • Resource managers • G8/S8 • Acquisition professionals

- Design program data architecture: Program, Milestone, Resource, Risk Object Types
- Ingest milestone and budget data from GFEBS/GCSS-A/IMS via Pipeline Builder
- Build milestone tracking dashboards with RAG status and automated variance computation
- Visualize budget execution (obligation rates) and flag at-risk funding lines
- Conduct portfolio analysis across programs in Contour and Quiver
- Build automated reporting pipelines with scheduled refresh and failure alerting
- Produce commander-ready IPR/EAB products from MSS without manual PowerPoint updates
- Apply access management, classification, and data stewardship for PM applications

PREREQ: TM-30 • ADVANCED: TM-50J

TM-40K — Knowledge Manager Track

TM-30 REQUIRED

Knowledge Repositories, AIP Summarization & Lessons Learned

KMOs • 37F • S2/S3 KM roles • AAR facilitators

- Design knowledge architecture for AAR, lessons learned, doctrine, and SOP repositories
- Build AAR capture systems using Workshop forms and Object Type pipelines
- Design and operate lessons-learned ingestion and tagging pipelines
- Use AIP Logic for knowledge summarization, search augmentation, and theme extraction
- Build full-text and semantic search systems over knowledge repositories
- Manage doctrine and SOP version control within Foundry
- Build personnel expertise mapping (skills/experience registries)
- Design knowledge transfer and unit continuity processes using MSS

PREREQ: TM-30 • ADVANCED: TM-50K

TM-40L — Software Engineer Track

TM-30 REQUIRED

OSDK, Full-Stack Foundry Applications & Platform SDK

SWEs • 17A/17C • 25D

- Authenticate and query the Foundry Ontology via OSDK (TypeScript/Python)
- Execute Actions, subscribe to Object changes, handle pagination and filtering via OSDK
- Use Foundry Platform SDK for dataset operations, file management, and branch management
- Build TypeScript Functions on Objects (computed properties, bulk query patterns)
- Write and test complex Action validators with TypeScript
- Build Slate applications integrated with the Foundry API
- Apply USAREUR-AF code review and deployment standards for MSS applications

PREREQ: TM-30 • ADVANCED: TM-50L

▼ WARFIGHTING FUNCTION TRACKS (TM-40A–F)

DESIGNATION	TRACK	PUBLICATION
TM-40A	Intelligence	TM_40A_INTELLIGENCE.pdf
TM-40B	Fires	TM_40B_FIRES.pdf
TM-40C	Movement & Maneuver	TM_40C_MOVEMENT_MANEUVER.pdf
TM-40D	Sustainment	TM_40D_SUSTAINMENT.pdf
TM-40E	Protection	TM_40E_PROTECTION.pdf

DESIGNATION	TRACK	PUBLICATION
TM-40F	Mission Command	TM_40F_MISSION_COMMAND.pdf

TM-50

Advanced Developer Tracks

Prerequisite: TM-40 (by track) • Tracks TM-50G–L • All 6 tracks available

▶ BLUF — ADVANCED TRACKS

The TM-50 series provides advanced-level instruction for each developer track, building directly on the corresponding TM-40 specialist manual (TM-40G–L). Intended for senior technical leads, platform architects, and senior developers designing enterprise-scale MSS capabilities. All 6 tracks are complete and available.

TM-50 SERIES — PUBLICATIONS

TM-50G — ORSA Advanced

ADVANCED

Advanced ORSA

Prerequisite: TM-40G

- Nonlinear programming, stochastic models
- Agent-based modeling (ABMS)
- Campaign wargame data architecture

TM-50H — AI Engineer Advanced

ADVANCED

Advanced AI Engineering

Prerequisite: TM-40H

- Multi-agent orchestration & shared state
- Advanced RAG, domain-adapted LLMs
- AI red-team assessment & observability

TM-50I — ML Engineer Advanced

ADVANCED

Advanced ML Engineering

Prerequisite: TM-40I

- Automated retraining pipelines

- Transformer fine-tuning, GNNs
- Federated retraining, adversarial robustness

TM-50J — PM Advanced

ADVANCED

Advanced Program Management

Prerequisite: TM-40J

- PI planning, cross-team governance
- GO/SES briefing, Palantir partnership
- Technical debt at program scale

TM-50K — KM Advanced

ADVANCED

Advanced Knowledge Management

Prerequisite: TM-40K

- Federated KM architecture, NATO integration
- STANAG 4778 conformance
- Knowledge graphs at scale

TM-50L — SWE Advanced

ADVANCED

Advanced Software Engineering

Prerequisite: TM-40L

- Scale, multi-tenancy, event streaming
- OWASP, SAST, authorized pen testing
- Architecture review, platform governance

• NOTE — PREREQUISITES

Each TM-50 track requires completion of the corresponding TM-40 track. Personnel should confirm TM-40 proficiency with their data steward before beginning TM-50 content. Contact the USAREUR-AF Operational Data Team for access questions.

DOCTRINE

Data Literacy Publications

Foundational doctrine • Platform-agnostic

▶ BLUF

Before touching MSS, understand *why* data matters. Data Literacy for Senior Leaders is written for commanders and senior leaders. Data Literacy Technical Reference is the comprehensive reference for all personnel. Neither publication is platform-specific.

SENIOR LEADERS (O-5+ / SGM+)

Data Literacy for Senior Leaders

Written for commanders, senior NCOs, and senior civilians. Covers command responsibilities, evaluating data products, directing a data-capable formation, and decision frameworks.

Format: Short (~20–30pp). Principles, not procedures. Chapter/paragraph numbered.

Key topics: Commander's data responsibilities • Evaluating analytical products • Standing up MSS capability • Data governance and stewardship

ALL PERSONNEL

Data Literacy Reference

Comprehensive platform-agnostic data literacy reference. Recommended prior reading before TM-10.

Format: Long (~50–100pp). Examples, vignettes, detailed explanations, annexes.

Key topics: Data types and structures • Pipeline concepts • Data quality • Analysis fundamentals • Security and classification • Operational data integration • Governance

FULL PUBLICATIONS INDEX

PUBLICATION	AUDIENCE	PURPOSE	WHEN TO READ
Data Literacy (SL)	O-5+ / SGM+, Sr Civilians	Principles, command responsibilities	Before directing MSS use
Data Literacy	All personnel	Comprehensive data literacy reference	Before TM-10 (recommended)

PUBLICATION	AUDIENCE	PURPOSE	WHEN TO READ
TM-10	All personnel	Operate MSS as end user	Before first MSS access
TM-20	All staff	Build pipelines, Ontology, Workshop via UI — no code	After TM-10
TM-30	Data-adjacent specialists	Design complex apps; governance; C2DAO standards	After TM-10 + TM-20
TM-40G	ORSA	Statistical modeling, simulation, wargame analytics	After TM-30
TM-40H	AI Engineers	AIP Logic authoring, Agent Studio, LLM integration	After TM-30
TM-40I	ML Engineers	Code Workspaces, model training, MLOps	After TM-30
TM-40J	PMs / G8	PM dashboards, milestone tracking, portfolio analysis	After TM-30
TM-40K	KMs / KMOs	Knowledge repositories, AIP summarization, lessons learned	After TM-30
TM-40L	SWEs	OSDK, full-stack Foundry apps, TypeScript Functions	After TM-30
TM-50G–L	Senior developers (all tracks)	Advanced versions of each TM-40 specialist track	After TM-40 (by track)

CORE DATA LITERACY CONCEPTS

DATA TYPES

Structured (tables, rows, columns) • Semi-structured (JSON, XML) • Unstructured

DATA PIPELINE

Raw → Staging → Curated → Ontology → Application. Each layer adds quality,

(documents, images). MSS ingests all three types from Army source systems.

structure, and meaning. Never modify raw data — it is the source of truth.

DATA QUALITY

Accuracy, completeness, consistency, timeliness, uniqueness. Bad decisions follow bad data. Report quality issues to your data steward rather than working around them.

DATA GOVERNANCE

Ownership, stewardship, access control, classification markings. Every dataset has an owner. Every user has a role. Access is audited. Misuse is tracked and prosecuted.

QUICK REF

MAVEN SMART SYSTEM — QUICK REFERENCE CARD

i NEW TO MSS?

Start with the Quick Start guide before reading TM-10. Operational in 30 minutes: log in via CAC, navigate to your unit's app, filter data, export a view. → **QUICK_START.pdf** • No account yet? Contact your unit data steward.

▶ BLUF

All personnel start at TM-10. Builders add TM-20; data-adjacent specialists continue to TM-30; technical roles select their TM-40 track below.

FIND YOUR MANUAL

YOU ARE…	START HERE	OPEN IT
Any personnel — viewing data for the first time	TM-10 — Maven User	TM-10 →
Staff building dashboards or simple pipelines (no coding)	TM-20 — Builder	TM-20 →
Data steward, frequent builder, or data-adjacent role (17/25-series, G2, analyst)	TM-30 — Advanced Builder	TM-30 →
Technical specialist or warfighting function developer — ORSA • AI/ML Eng • PM • KM • SWE	TM-40 Specialist Track	TM-40 ▼
Advanced specialist — post TM-40 completion	TM-50 Advanced Track	TM-50 ▼
O-5 / SGM+ — directing a data-capable formation	Data Literacy for Senior Leaders	Doctrine →
All personnel — foundational data concepts before TM-10	Data Literacy Technical Reference	PDF →
Anyone — unfamiliar term or concept	Glossary — Data & Foundry	Doctrine →

COMMON DAILY TASKS

TASK	HOW
Find a record	Search bar or Filter Panel
Filter the view	Select values in the Filter Panel
Export data	**Export** / **Download** button → CSV or Excel

SECURITY — DO NOT

- Do **not** export data to a personal device or unapproved storage
- Do **not** share your MSS **credentials** with anyone — URLs and screenshots are fine unless data is sensitive
- Do **not** enter classified information into MSS unless your instance is approved for that classification level

TASK	HOW
Submit or update a record	Click record → **Action button** → fill in → Submit

- Do **not** screenshot or share MSS screens containing data above your network's approved classification
- Do **not** use MSS on public or unsecured Wi-Fi
- If you see data you should not have access to — **stop and report to your data steward immediately**

WHEN IT BREAKS

Can't log in
Check CAC is fully inserted; try a different port. No account? Contact your unit data steward to request provisioning.

Button greyed out / no access
You're missing a role or write permission. Contact your data steward to request the correct access level.

App won't load
Hard-refresh (Ctrl+Shift+R). Clear cache. Try a different browser. Still broken → **Help Desk →**

DATA STEWARD
Account access • data issues • permission requests

HELP DESK
App broken • won't load • technical errors → **Support →**

ODT
Training questions • new app or pipeline requests

SUPPORT

Getting Help — USAREUR-AF

▶ BLUF

Know who to call before you have a problem. Route issues correctly from the start. Collect information before calling for help — it speeds resolution significantly.

⚠ WARNING — SECURITY INCIDENTS

If you suspect a security violation, report immediately to your supervisor and unit security officer. Do not investigate or resolve it yourself. Preserve the screen state; do not close the window or clear the browser.

CONTACT ROUTING

ISSUE	ROUTE TO	PRIORITY
Cannot log in / CAC issues	MSS Help Desk	Normal
No access to a project or dataset	Unit data steward	Normal
Data appears incorrect	Unit data steward (do not self-correct)	Normal
System error, crash, or outage	MSS Help Desk + screenshot + error code	Normal
Application broken or not loading	MSS Help Desk	Normal
Building question / how-to	Unit data lead or USAREUR-AF data team	Normal
Governance / C2DAO question	USAREUR-AF C2DAO	Normal
Security incident	**Supervisor + unit security officer**	**IMMEDIATE**

BEFORE CALLING FOR HELP — COLLECT THIS INFORMATION

- Your username and unit
- Name of the application, dataset, or pipeline you were using
- Exact error message (screenshot preferred)
- Time the error occurred (local or Zulu — state which)

- Steps that led to the error in order
- Browser and workstation you are using

PREREQUISITES BEFORE FIRST LOGIN

1. **Annual Cyber Awareness Training** — required for all DoD personnel; must be current
2. **MSS User Onboarding Brief** — provided by unit data steward
3. **Account request approved** — submit through your unit data steward; allow 3–5 business days

USAREUR-AF DATA TEAM

LOCATION & HIGHER HQ

EUCOM Theater — Europe & Africa

Headquarters, United States Army Europe and Africa
USAREUR-AF Operational Data Team
Army AI/Data Accelerator (C2DAO)

PUBLICATIONS

All training publications are maintained by the USAREUR-AF Operational Data Team. Contact your unit data steward for access to source files and this application. Version 3.0 — March 2026.

FEEDBACK & CORRECTIONS

Route corrections through your unit data steward to the USAREUR-AF Operational Data Team. Include: publication name, section, and description of the issue.

• NOTE — DISTRIBUTION

All publications in this curriculum are UNCLASSIFIED and approved for public release. Distribution is unlimited. Do not assume these publications apply to classified MSS configurations without consulting your data steward.

DOCUMENTS

All Training Publications

Click any publication to open the PDF

Doctrine & Reference

DATA LIT — SENIOR LEADERS

Data Literacy for Leaders

Senior leader version — strategic principles, decision frameworks. Commanders and O6+.

DATA LITERACY

Data Literacy — Comprehensive Reference

Full doctrine reference, examples, vignettes, and annexes. Platform-agnostic.

GLOSSARY

Data & Foundry Term Glossary

Combined data concepts and Foundry/MSS term equivalency. v1.3.

STANDARDS

Naming & Governance Standards

Codified naming conventions, governance checklists — 10 sections, 2 appendices.

QUICK REF

MSS Quick Reference Cheatsheet

At-a-glance reference card for common MSS tasks and patterns.

MTP

Mission Training Plan — MSS

TLOs, ELOs, Go/No-Go criteria, schedule templates. 961 lines.

Syllabi

SYLLABUS

TM-10 Syllabus

Maven User — all staff.

SYLLABUS

TM-20 Syllabus

No-Code Builder — all staff.

SYLLABUS

TM-30 Syllabus

Advanced No-Code Builder — data-adjacent MOS.

SYLLABUS

TM-40G Syllabus

ORSA specialists.

SYLLABUS

TM-40H Syllabus

AI Engineers.

SYLLABUS

TM-40I Syllabus

ML Engineers.

SYLLABUS

TM-40J Syllabus

Program Managers.

SYLLABUS

TM-40K Syllabus

Knowledge Managers.

SYLLABUS

TM-40L Syllabus

Software Engineers.

SYLLABUS

TM-40A Syllabus

Intelligence warfighting function track.

SYLLABUS

TM-40B Syllabus

Fires warfighting function track.

SYLLABUS

TM-40C Syllabus

Movement & Maneuver warfighting function track.

SYLLABUS

TM-40D Syllabus

SYLLABUS

TM-40E Syllabus

Sustainment warfighting function track.

Protection warfighting function track.

SYLLABUS

TM-40F Syllabus

Mission Command warfighting function track.

Technical Manuals — Foundation (All Staff)

TM-10

Maven User

Operate MSS, consume data, use dashboards and AIP. Required for all staff.

TM-20

No-Code Builder

Pipeline Builder, Workshop, basic Ontology. All staff with build responsibilities.

TM-30

Advanced No-Code Builder

Advanced Workshop, complex pipelines, AIP Logic. Data-adjacent MOS (17/25-series, G2).

Technical Manuals — Specialist Tracks (TM-40 Series)

TM-40G — ORSA

SPECIALIST

Operational Research Systems Analysis

ORSA specialists — quantitative methods, commander products

- Regression, classification, forecasting
- Monte Carlo COA analysis
- Optimization and sensitivity analysis

TM-40H — AI Engineer

SPECIALIST

AI Engineering

AI engineers — AIP Logic, Agents, LLM integration

- AIP Logic workflow design
- Agent configuration
- LLM integration patterns

TM-40I — ML Engineer

SPECIALIST

Machine Learning Engineering

ML engineers — model training, validation, deployment

- Feature engineering, experiment tracking
- Batch inference, model versioning
- Drift detection, retraining pipelines

TM-40J — Program Manager

SPECIALIST

Data Program Management

PMs — pipelines, milestones, portfolio health

- Scrum / Kanban for data projects
- ML/AI project lifecycle
- Risk register, release planning

TM-40K — Knowledge Manager

SPECIALIST

Knowledge Management

KMs — forms, lessons learned, institutional memory

- Knowledge ontology design
- Lessons-learned intake pipeline
- SOP review workflows

TM-40L — Software Engineer

SPECIALIST

Software Engineering

SWEs — Python/TypeScript, OSDK, code transforms

- OSDK & Platform SDK
- Functions on Objects, Actions
- CI/CD, security, Slate

Technical Manuals — Advanced Tracks (TM-50 Series)

TM-50G — ORSA Advanced

ADVANCED

Advanced ORSA

Nonlinear programming, stochastic models, ABMS, campaign assessment

TM-50H — AI Engineer Advanced

ADVANCED

Advanced AI Engineering

Multi-agent orchestration, RAG, red-team assessment, AI observability

TM-50I — ML Engineer Advanced

ADVANCED

Advanced ML Engineering

AutoML, Transformer fine-tuning, GNNs, federated retraining, adversarial robustness

TM-50J — PM Advanced

ADVANCED

Advanced Program Management

PI planning, cross-team governance, GO/SES briefing, Palantir partnership

TM-50K — KM Advanced

ADVANCED

Advanced Knowledge Management

Federated KM architecture, NATO integration, STANAG 4778, knowledge graphs

TM-50L — SWE Advanced ADVANCED

Advanced Software Engineering

Scale, multi-tenancy, Kafka, OWASP, SAST, architecture review, platform governance

Warfighting Function Tracks (TM-40A–F)

TM-40A — INTELLIGENCE

Intelligence Warfighting Function

MSS integration for intelligence operations and data products.

TM-40B — FIRES

Fires Warfighting Function

MSS integration for fires coordination and targeting data.

TM-40C — MOVEMENT & MANEUVER

Movement & Maneuver WFF

MSS integration for maneuver and mobility data products.

TM-40D — SUSTAINMENT

Sustainment Warfighting Function

MSS integration for logistics and sustainment data.

TM-40E — PROTECTION

Protection Warfighting Function

MSS integration for force protection data and reporting.

TM-40F — MISSION COMMAND

Mission Command WFF

MSS integration for mission command, C2, and operational reporting.

Concepts Guides

TM-40G Concepts Guide

ORSA

Key concepts and terminology for the ORSA specialist track.

TM-40H Concepts Guide

AI ENGINEER

Key concepts and terminology for the AI Engineer track.

TM-40I Concepts Guide

ML ENGINEER

Key concepts and terminology for the ML Engineer track.

TM-40J Concepts Guide

PROGRAM MGR

Key concepts and terminology for the Program Manager track.

TM-40K Concepts Guide

KNOWLEDGE MGR

Key concepts and terminology for the Knowledge Manager track.

TM-40L Concepts Guide

SOFTWARE ENG

Key concepts and terminology for the Software Engineer track.

TM-50G Concepts Guide

ORSA ADV

Advanced concepts and terminology for TM-50G.

TM-50H Concepts Guide

AI ENG ADV

Advanced concepts and terminology for TM-50H.

TM-50I Concepts Guide

MLE ADV

Advanced concepts and terminology for TM-50I.

TM-50J Concepts Guide

PM ADV

Advanced concepts and terminology for TM-50J.

TM-50K Concepts Guide

KM ADV

Advanced concepts and terminology for TM-50K.

TM-50L Concepts Guide

SWE ADV

Advanced concepts and terminology for TM-50L.

Practical Exercises & Assessments

EXERCISE

TM-10 Practical Exercise

EXERCISE

TM-20 Practical Exercise

Operator basics — hands-on exercise.

No-code builder — hands-on exercise.

EXERCISE

TM-30 Practical Exercise

Advanced builder — hands-on exercise.

EXERCISE

TM-40G Exercise (ORSA)

ORSA specialist practical exercise.

EXERCISE

TM-40H Exercise (AI Eng)

AI Engineer practical exercise.

EXERCISE

TM-40I Exercise (ML Eng)

ML Engineer practical exercise.

EXERCISE

TM-40J Exercise (PM)

Program Manager practical exercise.

EXERCISE

TM-40K Exercise (KM)

Knowledge Manager practical exercise.

EXERCISE

TM-40L Exercise (SWE)

Software Engineer practical exercise.

Administrative & Institutional Documents

POI

Program of Instruction

MSS curriculum Program of Instruction.

TEO

Training & Evaluation Outline

MSS training and evaluation outline.

SOP

Enrollment SOP

Standard operating procedure for course enrollment.

SOP

Curriculum Maintenance SOP

Procedures for maintaining and updating curriculum.

SCHEDULE

Annual Training Schedule

Full-year MSS training schedule.

POLICY

Policy Letter

Command policy letter governing MSS training.

TEMPLATE

Completion Certificate

Course completion certificate template.

TEMPLATE

AAR Template

After Action Review template for training events.

PLAN

Faculty Development Plan

Instructor and facilitator development plan.

CAD

Course Administrative Data

Course administrative data sheet.

Lesson Plans

LESSON PLAN

TM-10 Lesson Plans

Instructor lesson plans for TM-10 Maven User.

LESSON PLAN

TM-20 Lesson Plan Outlines

Instructor lesson plan outlines for TM-20 Builder.

LESSON PLAN

TM-30 Lesson Plan Outlines

Instructor lesson plan outlines for TM-30 Advanced Builder.

LESSON PLAN

TM-40 Specialist Lesson Plans

Instructor lesson plan outlines for all TM-40 specialist tracks.

TEMPLATE

Lesson Plan Template

Blank lesson plan template for course development.

TASK INDEX

What Do You Want To Do?

Search • Filter by category • Click badge to open PDF

UNCLASSIFIED

MSS REF

MSS TASK REFERENCE INDEX

USAREUR-AF Operational Data Team | What do you want to do? Search below.

SEARCH: Type a task, keyword, or topic...

397 / 256

FILTER: All | Setup & Access | Navigation | Data & Analysis | No-Code Building | Ontology & Modeling | AIP & AI | Code & SDK | Machine Learning | Program Mgmt | Knowledge Mgmt | Governance & Security | Advanced / Expert

SETUP & ACCESS

Set up account before first login	TM-10	Ch 2
Log in to MSS with CAC	TM-10	Ch 2
Log out of MSS	TM-10	Ch 2
Verify your own markings and access level	TM-10	Ch 6
Configure new ORSA Code Workspace	TM-40A	Ch 2
Create and configure Code Workspace	TM-40C	Ch 2
Manage packages in Code Workspace	TM-40A, TM-40C	Ch 2
Manage compute and environment resources	TM-40C	

SCHEDULE

Upcoming Training — TM-10 / TM-20 / TM-30

Foundation courses • Contact POC to enroll

MSS FOUNDATION TRAINING — UPCOMING SCHEDULE

TM-10 • TM-20 • TM-30 • USAREUR-AF ODT

TM-10 Maven User Manual — All Personnel

2 days (16 hrs) • No prereq

DATES	LOCATION	FORMAT	POC	SEATS AVAIL.	STATUS
14–15 APR 2026	Wiesbaden, Clay Kaserne, Bldg 3312, Rm 104	In-Person	SSG Johnson	20 / 20	OPEN
05–06 MAY 2026	Grafenwöhr, Bldg 244, Conf Rm B	In-Person	SFC Davis	8 / 20	8 REMAINI
18–19 JUN 2026	Stuttgart, Kelley Bks, Bldg 3357	In-Person	SSG Martinez	20 / 20	OPEN
09–10 JUL 2026	Virtual (MS Teams)	Virtual	SSG Johnson	30 / 30	OPEN

Enrollment: Contact listed POC to reserve a seat. Ensure your MSS account request is submitted at least 5 business days before course start. Virtual sessions require MS Teams access and a headset.

TM-20 No-Code Builder Manual — All Staff

3 days (24 hrs) • Prereq: TM-10

DATES	LOCATION	FORMAT	POC	SEATS AVAIL.	STATUS
21–23 APR 2026	Wiesbaden, Clay Kaserne, Bldg 3312, Rm 104	In-Person	SFC Chen	7 / 15	7 REMAINI
12–14 MAY 2026	Grafenwöhr, Bldg 244, Conf Rm B	In-Person	SSG Williams	3 / 15	3 REMAINI

USAREUR-AF OPERATIONAL DATA TEAM • TM-10 • TM-20 • TM-30 • TM-40A–L • TM-50G–L • Data Lit (SL) • Data Literacy

VERSION 3.1 — MARCH 2026 — DIST: APPROVED FOR PUBLIC RELEASE — DISTRIBUTION UNLIMITED

UNCLASSIFIED